

Pins

These pull-ups are only for hall effect sensors that specify.

The "NC/0u1F" capacitors are capacitors that should only be placed if you are unable to place them on the hall effect sensors themselves. This is not needed for motors with built-in hall effect sensors.

Step-Down Converter

Figure 15. Step-down converter

Voltage Regulator

The diagram illustrates a voltage regulator circuit. The input voltage (VIN) is 5V, and the output voltage (VOUT) is 3.3V. The regulator is an AMS1117-3V3. The input is connected to a 5V supply (VCC_5V) through a 10uF capacitor (C25). The output is connected to a 3.3V supply (VCC_3V3) through a 10uF capacitor (C27) and a 180 ohm resistor (R24). The ground/adjust (GND/ADJ) pin is connected to ground through a 0.1uF capacitor (C26). The output is also connected to a red LED (D9) through a 180 ohm resistor (R24).

MCU Buttons

Default States

The diagram illustrates the default states for a 74VHC00 NAND gate. The inputs A and B are connected to VCC_3V3 through resistors R26, R27, R28, R29, and R30. Inputs EN, FAULT, SPE, SINGLE, and AMPL are connected to input A. Input B is connected to input A through resistor R31. The output Y is connected to input A through resistor R31 and to GND through resistor R30. The output Y is also connected to input B through resistor R31.

MOSFETs

The three diagrams illustrate different voltage divider configurations using MOSFETs to interface a variable voltage source (VM) with a microcontroller pin (pU, pV, pW). Each circuit uses two N-channel MOSFETs (U2, U3; U4, U5; U6, U7) and a 2u2F capacitor (C12, C14, C16) to interface a variable voltage source (VM) with a microcontroller pin (pU, pV, pW).

- Left Circuit (pU):** The input VM is connected to the gates of U2 and U3. The source of U2 is connected to the source of U3, which is connected to ground. The drain of U2 is connected to the microcontroller pin pU. The gate of U3 is connected to the drain of U2. The source of U3 is connected to ground. The drain of U3 is connected to ground. A 2u2F capacitor (C12) is connected between VM and ground. A 1nF/100V capacitor (C13) is connected between the gate of U3 and ground. A 33mF capacitor (R9) is connected between the gate of U2 and ground. A 2R2 resistor (R10) is connected between the gate of U2 and the drain of U2. A 2R2 resistor (R11) is connected between the gate of U3 and the drain of U3.
- Middle Circuit (pV):** The input VM is connected to the gates of U4 and U5. The source of U4 is connected to the source of U5, which is connected to ground. The drain of U4 is connected to the microcontroller pin pV. The gate of U5 is connected to the drain of U4. The source of U5 is connected to ground. The drain of U5 is connected to ground. A 2u2F capacitor (C14) is connected between VM and ground. A 1nF/100V capacitor (C15) is connected between the gate of U5 and ground. A 33mF capacitor (R12) is connected between the gate of U4 and ground. A 2R2 resistor (R13) is connected between the gate of U4 and the drain of U4. A 2R2 resistor (R14) is connected between the gate of U5 and the drain of U5.
- Right Circuit (pW):** The input VM is connected to the gates of U6 and U7. The source of U6 is connected to the source of U7, which is connected to ground. The drain of U6 is connected to the microcontroller pin pW. The gate of U7 is connected to the drain of U6. The source of U7 is connected to ground. The drain of U7 is connected to ground. A 2u2F capacitor (C16) is connected between VM and ground. A 1nF/100V capacitor (C17) is connected between the gate of U7 and ground. A 33mF capacitor (R15) is connected between the gate of U6 and ground. A 2R2 resistor (R16) is connected between the gate of U6 and the drain of U6. A 2R2 resistor (R17) is connected between the gate of U7 and the drain of U7.

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